CCNA LEARNING SERIES

Day 4: Subnetting Practice & VLSM

Block size • Network address • Host range • Broadcast • VLSM planning

Learning objective
Practise finding subnet details from an IPv4 address and prefix length, understand block size, and learn how VLSM allocates different-sized subnets efficiently.

In this document

1. Subnetting recap
2. Block-size method
3. Worked subnetting examples
4. Network, host range and broadcast
5. VLSM concepts and planning
6. Verification commands
7. Interview questions
8. Practice exercises with answers

A beginner-friendly practice guide with step-by-step calculations and revision questions.

1. Subnetting Recap

Subnetting divides one IP network into smaller logical networks called subnets. A prefix such as /26 tells us how many bits are used for the network portion.

Host bits = 32 – prefix length
Total addresses = $2^{\text{host bits}}$
Traditional usable hosts = total addresses – 2

The usual usable-host formula excludes the network and broadcast addresses. Special cases such as /31 point-to-point links and /32 host routes are exceptions.

2. Understanding Block Size

For common subnetting questions, block size helps identify where each subnet starts. In the octet where the subnet mask is neither 255 nor 0: **Block size = 256** – **subnet-mask value in that octet.**

Prefix	Subnet mask	Block size	Usable hosts
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

For /26, block size = 256 – 192 = 64. Subnet boundaries are 0, 64, 128 and 192.

3. Worked Example: 192.168.1.70/26

Step 1: /26 mask = 255.255.255.192.
Step 2: Block size = 256 – 192 = 64.
Step 3: Boundaries are 0, 64, 128 and 192.
Step 4: 70 falls between 64 and 127.

Subnet detail	Answer
Network address	192.168.1.64
First usable host	192.168.1.65
Last usable host	192.168.1.126
Broadcast address	192.168.1.127
Total / usable addresses	64 / 62

4. More Worked Examples

Example A: Find subnet details for 10.1.5.140/27

/27 mask = 255.255.255.224. Block size = 256 – 224 = 32. Boundaries include 0, 32, 64, 96, 128, 160, 192 and 224. Since 140 lies between 128 and 159:

Subnet detail	Answer
Network address	10.1.5.128
First usable host	10.1.5.129
Last usable host	10.1.5.158
Broadcast address	10.1.5.159
Total / usable addresses	32 / 30

Example B: Find subnet details for 172.16.4.200/28

/28 mask = 255.255.255.240. Block size = 256 – 240 = 16. The address 200 lies in the 192–207 block:

Subnet detail	Answer
Network address	172.16.4.192
First usable host	172.16.4.193
Last usable host	172.16.4.206
Broadcast address	172.16.4.207
Total / usable addresses	16 / 14

5. What is VLSM?

VLSM stands for **Variable Length Subnet Mask**. It allows a network to be divided into subnets of different sizes according to the number of hosts needed. This can use address space more efficiently than assigning the same size to every subnet.

Key idea: Allocate the largest host requirement first, then allocate smaller subnet blocks from the remaining address space. Each subnet must start on a valid boundary for its prefix length.

6. VLSM Planning Example

A company has 192.168.10.0/24 and needs three LANs: 100 hosts, 50 hosts and 20 hosts. Choose the smallest traditional subnet that supports each requirement, then allocate largest first.

LAN need	Chosen subnet	Usable hosts	Network
100 hosts	/25	126	192.168.10.0/25
50 hosts	/26	62	192.168.10.128/26
20 hosts	/27	30	192.168.10.192/27

This uses 128 + 64 + 32 = 224 addresses. The remaining block is 192.168.10.224/27, available for future use.

7. VLSM Allocation Details

LAN	Network	Usable host range	Broadcast
100-host	192.168.10.0/25	192.168.10.1 – 126	192.168.10.127
50-host	192.168.10.128/26	192.168.10.129 – 190	192.168.10.191
20-host	192.168.10.192/27	192.168.10.193 – 222	192.168.10.223
Remaining	192.168.10.224/27	192.168.10.225 – 254	192.168.10.255

8. Verification and Troubleshooting Commands

Command	Purpose
ipconfig /all	View detailed IPv4 settings in Windows
ip addr	View interface addresses in Linux
ip route	View routing table in Linux
ping <IP-address>	Test reachability to a destination
tracert <IP-address>	Trace route in Windows
show ip interface brief	Quickly view Cisco interface status and IP addresses
show running-config	View active Cisco device configuration

On Cisco IOS, use show commands in the appropriate privileged EXEC mode. Commands and output can vary by device and software version.

9. Interview Questions and Answers

Q1. What is subnetting?

Subnetting divides a larger IP network into smaller logical networks.

Q2. What is block size?

It is the increment between subnet network addresses in the relevant octet. Calculate 256 minus the subnet-mask value in that octet.

Q3. What is the block size for /26?

The mask is 255.255.255.192, so block size = 256 – 192 = 64.

Q4. How many traditionally usable hosts are in /27?

There are 5 host bits, 32 total addresses and 30 traditionally usable host addresses.

Q5. What is VLSM?

Variable Length Subnet Masking allows different subnet sizes within the same larger network.

Q6. Why allocate the largest subnet first?

It places the largest required address block first and reduces the chance of leaving unusable gaps.

Q7. What is the broadcast address of 192.168.1.70/26?

192.168.1.127.

Q8. What is the network address of 10.1.5.140/27?

10.1.5.128.

Q9. Which Cisco command summarizes interface IP addresses and status?

show ip interface brief.

Q10. Difference between FLSM and VLSM?

FLSM uses the same subnet mask size for all subnets; VLSM allows different subnet mask sizes.

10. Practice Questions

Try these before checking the answers.

1. Find the block size for /27.
2. Find the network address of 192.168.2.145/26.
3. Find the broadcast address of 192.168.2.145/26.
4. How many traditionally usable hosts are in /28?
5. Find the network address of 192.168.5.50/27.
6. Find the broadcast address of 192.168.5.50/27.
7. What does VLSM stand for?
8. Which host requirement should normally be allocated first in VLSM?

Practice Answers

1. 32
2. 192.168.2.128
3. 192.168.2.191
4. 14 usable hosts
5. 192.168.5.32
6. 192.168.5.63
7. Variable Length Subnet Mask
8. The largest host requirement

Quick Revision

Remember	Answer
Block size	256 – subnet mask value in relevant octet
/26	64 total addresses; 62 traditionally usable hosts
/27	32 total addresses; 30 traditionally usable hosts
/28	16 total addresses; 14 traditionally usable hosts
VLSM	Different subnet sizes in one address block
VLSM order	Allocate largest subnet first
Cisco interface summary	show ip interface brief

Study tip: Practise five subnetting questions daily. For every IP/prefix, write the subnet mask, block size, network address, first host, last host and broadcast address.

End of Day 4 | Next topic: VLANs and Trunking